

Aditya Kulkarni

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Raleigh, NC -- Open to Palo Alto

PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) with production experience building agentic runtime infrastructure: Temporal.io session lifecycle orchestration, multi-model LLM integration (OpenAI, Anthropic, Google Gemini) via MCP tool execution, and Kafka-driven fault-tolerant state synchronization at 2,000 async inference requests/second. At Isomer AI, delivered 10+ backend features and 394 commits across 7 production deployments using Temporal workflows, Redis caching, and real-time event pipelines. IEEE-published researcher. Open to relocating to Palo Alto.

CORE COMPETENCIES

Agentic Runtime Systems

Temporal.io Workflow Orchestration

Multi-Model LLM Integration

Kafka / Event-Driven Architecture

Fault-Tolerant Distributed Systems

MCP Tool Execution

Redis Session State & Caching

Cloud-Native Deployment (GCP, Kubernetes)

WORK EXPERIENCE

Isomer AI May 2025 -- Present

Product Engineer Intern · Remote

- Designed **multi-model agentic tooling infrastructure** across 3 LLM providers (Anthropic Claude, OpenAI GPT-4o, Google Gemini) with **MCP-based tool execution** -- custom retry logic, fault tolerance, and cost monitoring across 8+ models in production
- Built **Temporal.io workflow orchestration** for AI-powered claims automation with **Redis session state**, real-time event pipelines (Pusher), and role-based access control across 7 production deployments
- Shipped 10+ backend features (394 commits) using TypeScript, Node.js, Express, PostgreSQL, Prisma, Docker, Auth0, and Sentry across a cloud-native stack
- Implemented document processing pipelines: PDF generation, email parsing via Microsoft Graph API, and S3 file storage integration

NC State University April 2025 -- Present

Graduate Research Assistant · Raleigh, NC

- Architected a **cloud-native genomic data platform** on Google Cloud Run processing 500GB datasets for 100+ researchers worldwide using JBrowse 2 with containerized microservice deployment
- Optimized ML model architecture via hyperparameter tuning and time-series analysis, achieving +12% accuracy improvement

Mphasis (HP Company) Jun 2024 -- Aug 2024

Software Developer Intern · India

- Built 5+ AI-driven automation workflows including OCR/NLP/ML health claim processor (+25% accuracy); deployed on AWS and GCP with Grafana observability dashboards and model red-teaming

Capgemini Jun 2023 -- Aug 2023

AI/ML Intern · India

- Designed **Kafka streaming pipelines** for real-time data ingestion (~20% latency) and Hadoop + Spark distributed batch processing (~15% processing time) for support ticket analysis at scale

PROJECTS

AlgoPulse -- Agentic AI Platform Independent 2025

Built an **agentic runtime platform** handling **2,000 async AI inference requests/second**. Identified API-polling bottleneck under load and re-architected to **Temporal.io session lifecycle orchestration**, **Kafka event-driven state synchronization**, and Supabase Edge Functions streaming AI reasoning loops directly to clients -- zero state loss under peak load.

TypeScript · Node.js · React · Temporal.io · Kafka · Supabase Edge Functions · Redis

Glance AI -- Privacy-First RAG System Independent 2025

Engineered a compliance-ready RAG pipeline for fintech/healthcare environments with per-tenant data isolation at the vector retrieval layer. Deterministic output constraints for hallucination mitigation in regulated contexts.

Python · LLMs · Vector Databases · RAG Architecture

PerCent -- GenAI Finance App HackNC24 -- Top 10 / 100+ teams

Built in 48 hours: LLM-powered financial assistant with RAG-based chatbot (Qwen-14B-Chat via Hugging Face), Selenium-driven data ingestion, and goal/expense tracking. Reduced manual research time by 30%.

Python · Hugging Face · Qwen-14B-Chat · RAG · Selenium

Mapify -- NLP to Structured Knowledge IEEE Published

T5-transformer pipeline converting 1,500+ PDFs and images into structured mind maps and flowcharts using NLTK and NetworkX. Reduced academic review time by 60%. Presented to 500+ attendees at IEEE conference.

T5-Transformer · NLTK · NetworkX · Python

EDUCATION

NC State University -- MS Computer Science Aug 2024 -- May 2026

Raleigh, NC · Relevant coursework: Artificial Intelligence, Operating Systems, Data Science, Algorithms, Software Engineering, Object-Oriented Development

Mumbai University -- BTech Computer Science Aug 2020 -- May 2024

CGPA: 9.32/10 · Coursework: Machine Learning, Big Data Infrastructure, Data Structures & Algorithms, OS, Software Engineering

AWARDS & RECOGNITION

- HackNC24 Finalist:** Top 10 of 100+ teams -- GenAI finance app with RAG and LLM integration
- IEEE Publication:** Mapify NLP pipeline presented to 500+ attendees at international conference
- National Hackathon Winner:** Top 10 in 6+ national hackathons; 3 complete MVPs in 48-hour builds
- Google DSC Lead:** Led 4+ tech events (hackathons, workshops) reaching 500+ students

SKILLS

- Languages:** Python, TypeScript, JavaScript, Java, SQL, C, R
- AI/ML & LLMs:** OpenAI API, Anthropic Claude, Google Gemini, LangChain, MCP, Hugging Face, TensorFlow, PyTorch, RAG, Pandas, NumPy
- Distributed Systems:** Kafka, Temporal.io, Apache Spark, Hadoop, Redis, gRPC, WebSockets
- Cloud & DevOps:** GCP, AWS, Docker, Kubernetes, Terraform, Grafana, Sentry, CI/CD
- Frameworks:** Node.js, React, Express, Prisma, MERN Stack, Tailwind, Pusher, Playwright